CHAPTER 4

ANATOMY OF FLOWERING PLANTS

CHAPTER ESSENCE AT A GLANCE:
Internal cellular specialization of epidermal, ground, and vascular tissue systems organizing primary plant organs.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Meristematic and Permanent Tissues:** Apical meristems drive primary vertical growth at root and shoot apices, while intercalary meristems regenerate grazed grasses. Simple permanent tissues comprise parenchyma (storage, photosynthesis), collenchyma (pectin corner thickenings providing flexible mechanical support in young stems), and sclerenchyma (thick lignified secondary walls in fibers and sclereids providing rigid support).

• **Epidermal Apparatus and Guard Cells:** The epidermis is coated with waxy cutin preventing transpirational water loss. Stomatal apparatus consists of a stomatal aperture flanked by twin guard cells: bean-shaped or kidney-shaped in dicots with thick inner and thin outer walls; dumbbell-shaped in monocot grasses with bulbous ends, surrounded by specialized subsidiary cells.

• **Dicot versus Monocot Stem Architecture:** Dicot stems feature a collenchymatous hypodermis, starch sheath endodermis, parenchymatous medullary rays, and a characteristic ring of conjoint, collateral, and open vascular bundles with intrafascicular cambium. Monocot stems possess a sclerenchymatous hypodermis and numerous scattered closed vascular bundles surrounded by sclerenchymatous bundle sheaths without cambium.

• **Dicot versus Monocot Root Architecture:** Roots feature an unthickened epiblema with unicellular root hairs and an endodermis with suberized Casparian strips regulating symplastic radial water transport. Dicot roots have diarch to hexarch radial xylem bundles with small or indistinct pith; Monocot roots have polyarch (more than six) radial xylem bundles with a large, well-developed pith.

ESSENTIAL TAXONOMICAL FRAMEWORKS & PROCESSES

$$\text{Dicot Stem Bundles} = \text{Ring Arrangement} + \text{Conjoint Collateral Open (Cambium Present)}$$

Vascular Organization in Dicotyledonous Stems

$$\text{Monocot Stem Bundles} = \text{Scattered Distribution} + \text{Closed (No Cambium)} + \text{Bundle Sheath}$$

Atactostelic Vascular Layout in Monocotyledonous Stems

$$\text{Root Xylem Arche} = \frac{\text{Dicot Root: Diarch to Hexarch (Small Pith)}}{\text{Monocot Root: Polyarch (Large Pith)}}$$

Radial Vascular Bundle Count Comparison in Roots

$$\text{Casparian Strip} = \text{Endodermal Radial Walls Suberized} \rightarrow \text{Obligate Symplastic Flow}$$

Endodermal Hydrophobic Barrier Regulating Water Uptake

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Remember that Secondary Growth topics (vascular cambium and cork cambium) were completely deleted under the rationalized syllabus.

• **Alert:** Guard cells are kidney-shaped in dicot leaves, but dumbbell-shaped in monocot grass leaves.

• **Alert:** Vascular bundles in dicot stems are open (cambium present between xylem and phloem), whereas in monocot stems they are closed (cambium absent).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between open vascular bundles and closed vascular bundles with an anatomical example for each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Open vascular bundles: Possess a strip of intrafascicular cambium between xylem and phloem, possessing the capacity to form secondary tissues (e.g., Dicotyledonous stems). **[1 Mark]**

• Closed vascular bundles: Lack cambium between xylem and phloem, incapable of producing secondary tissues (e.g., Monocotyledonous stems). **[1 Mark]**

2. What are bulliform cells? Explain their physiological role in the rolling and unrolling of grass leaves. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Bulliform cells are large, empty, colorless specialized cells located on the adaxial (upper) epidermis of monocot grass leaves. **[1 Mark]**

• When bulliform cells absorb water and become turgid, the leaf surface is exposed flat; during water stress, they become flaccid, causing the leaves to curl inwards to minimize transpirational water loss. **[1 Mark]**

3. Describe the anatomical structure of the endodermis in roots. Explain the biochemical nature and functional role of Casparian strips. [Score: 3 Marks]

Answer & Key Valuation Steps:

• The endodermis is the innermost layer of the root cortex, consisting of a single layer of barrel-shaped cells without intercellular spaces. **[1 Mark]**

• Casparian strips: Tangential and radial walls of endodermal cells possess a deposition of water-impermeable, waxy suberin material termed Casparian strips. **[1 Mark]**

• Functional role: The hydrophobic suberin blocks apoplastic water movement, forcing water and dissolved mineral solutes to cross the endodermal plasma membrane into the symplast toward the vascular cylinder. **[1 Mark]**

4. Tabulate three distinct anatomical differences between the transverse sections of a Dicot stem and a Monocot stem. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Hypodermis: Dicot stem possesses a collenchymatous hypodermis; Monocot stem possesses a sclerenchymatous hypodermis. **[1 Mark]**

• Vascular bundle arrangement: Dicot stem vascular bundles are arranged in a distinct concentric ring; Monocot stem vascular bundles are scattered throughout ground tissue (atactostele). **[1 Mark]**

• Cambium and sheath: Dicot vascular bundles are wedge-shaped, open with cambium, and lack bundle sheaths; Monocot vascular bundles are closed, oval, and enclosed by sclerenchymatous bundle sheaths. **[1 Mark]**

5. Compare a Dicot root with a Monocot root with respect to: (a) Number of xylem bundles, (b) Development of pith, and (c) Presence of cambium during secondary growth. [Score: 3 Marks]

Answer & Key Valuation Steps:

• (a) Number of xylem bundles: Dicot root has diarch to hexarch condition (2 to 6 vascular bundles); Monocot root has polyarch condition (more than 6 xylem bundles). **[1 Mark]**

• (b) Pith development: Pith is small, inconspicuous, or absent in dicot root; pith is large, conspicuous, and well-developed in monocot root. **[1 Mark]**

• (c) Cambium: Cambium develops as a secondary meristem in dicot root; cambium is completely absent in monocot root throughout life. **[1 Mark]**

6. Explain the structural organization of the stomatal apparatus, detailing the differences in guard cell morphology between dicots and grasses. [Score: 3 Marks]

Answer & Key Valuation Steps:

• The stomatal apparatus comprises the central stomatal aperture, twin flanking guard cells, and surrounding specialized subsidiary cells. **[1 Mark]**

• Dicot guard cells: Kidney-shaped or bean-shaped with highly thickened inelastic inner walls facing the pore and thin elastic outer walls, containing chloroplasts. **[1 Mark]**

• Monocot grass guard cells: Dumbbell-shaped with narrow middle portions and expanded bulbous ends, facilitating rapid opening and closing. **[1 Mark]**

7. A student examines an unknown plant cross-section under a microscope and records: (i) Collenchymatous hypodermis, (ii) Vascular bundles arranged in a ring, (iii) Conjoint, collateral, and open vascular bundles, (iv) Distinct central parenchymatous pith. Identify whether the section is a root or a stem, dicot or monocot, and justify each deduction. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Stem vs Root deduction: The section is a stem because vascular bundles are conjoint and collateral, hypodermis is present, and cuticular epidermis is evident (roots possess radial bundles and unthickened epiblema). **[1.5 Marks]**

• Dicot vs Monocot deduction: The section is a Dicotyledonous stem because vascular bundles are arranged in a regular ring and are open with cambium, supported by a collenchymatous hypodermis and distinct pith (monocot stems have scattered closed bundles and sclerenchymatous hypodermis). **[1.5 Marks]**

• Final identification: Primary transverse section of a Dicotyledonous stem (e.g., Sunflower stem). **[1 Mark]**